

YEAR 10 / INDUSTRIAL TECHNOLOGY METAL

BBQ and Case

STUDENT WORKBOOK



Name _____ Class _____

50 pages • A4 • 25 sheets duplex • Edition 1.0 • Course companion

Illustrative cover. Use teacher-issued project plans for construction.

NAVIGATION

Your learning and evidence route

Nine modules, four original drawing sheets and one connected folio.

Module	Read / check
1. Brief, evidence and materials	7–9
2. Drawings, rails and accurate setup	10–12
3. Joining and controlled assembly	13–15
4. WMS and repeatable work	16–18
5. Costs, flowcharts and terminology	19–21
6. Industry and environmental responsibility	22–24
7. Preparation and finishes	25–27
8. Case development and quality	28–30
9. Problem solving and final evidence	31–33

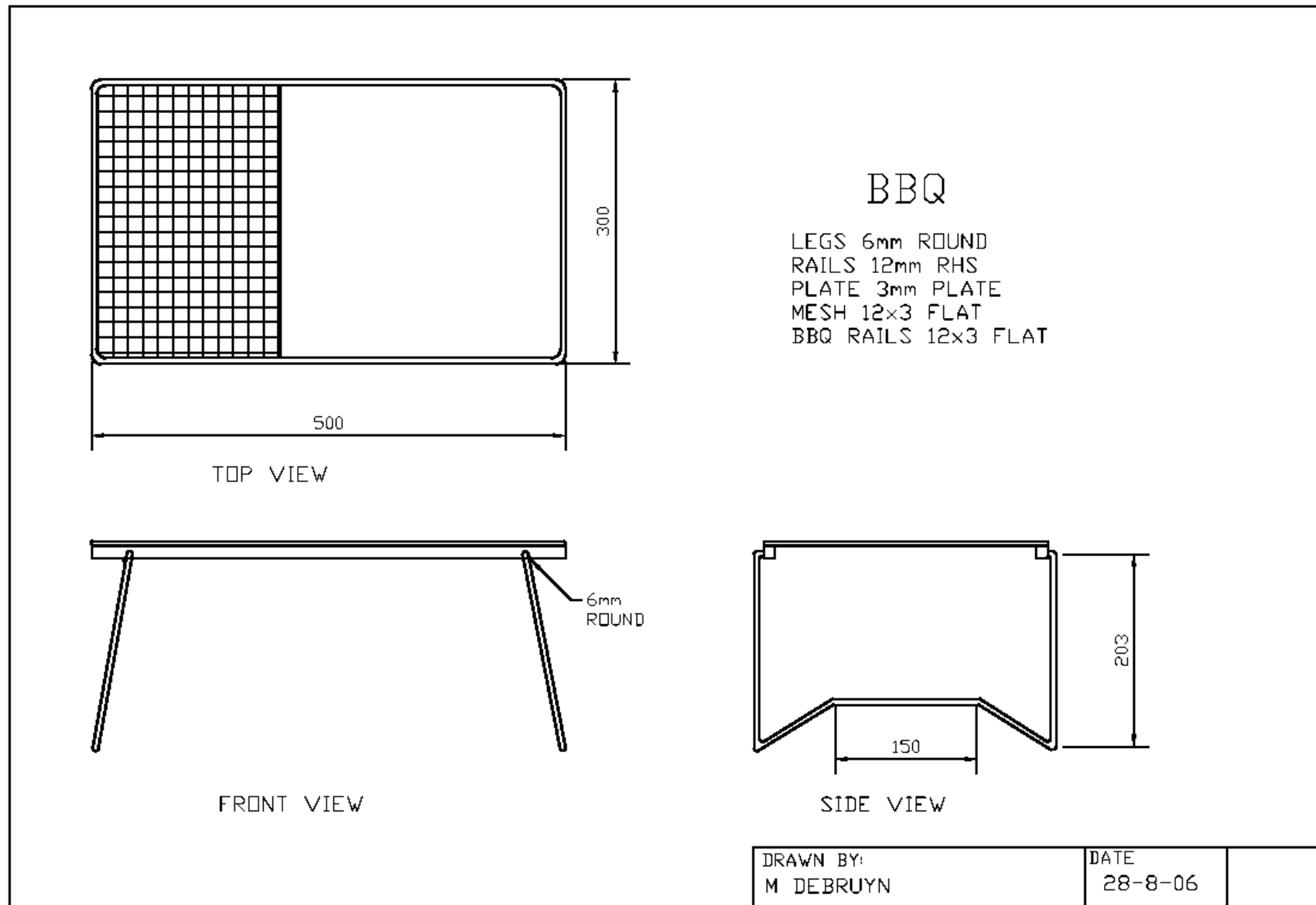
Original drawings: pages 3–6. Applied learning and studios: 34–38. Your 12-part evidence folio: 39–50. Exact written tasks are integrated with matching folio evidence; answer every printed prompt.

Current drawings, the brief, SOPs, WMS and teacher demonstrations control practical details. The four source plans are landscape A4 pages within this 50-page book. Print the PDF at actual size, preserve page orientation and use long-edge duplex: 25 sheets. Never scale a plan image for a missing dimension.

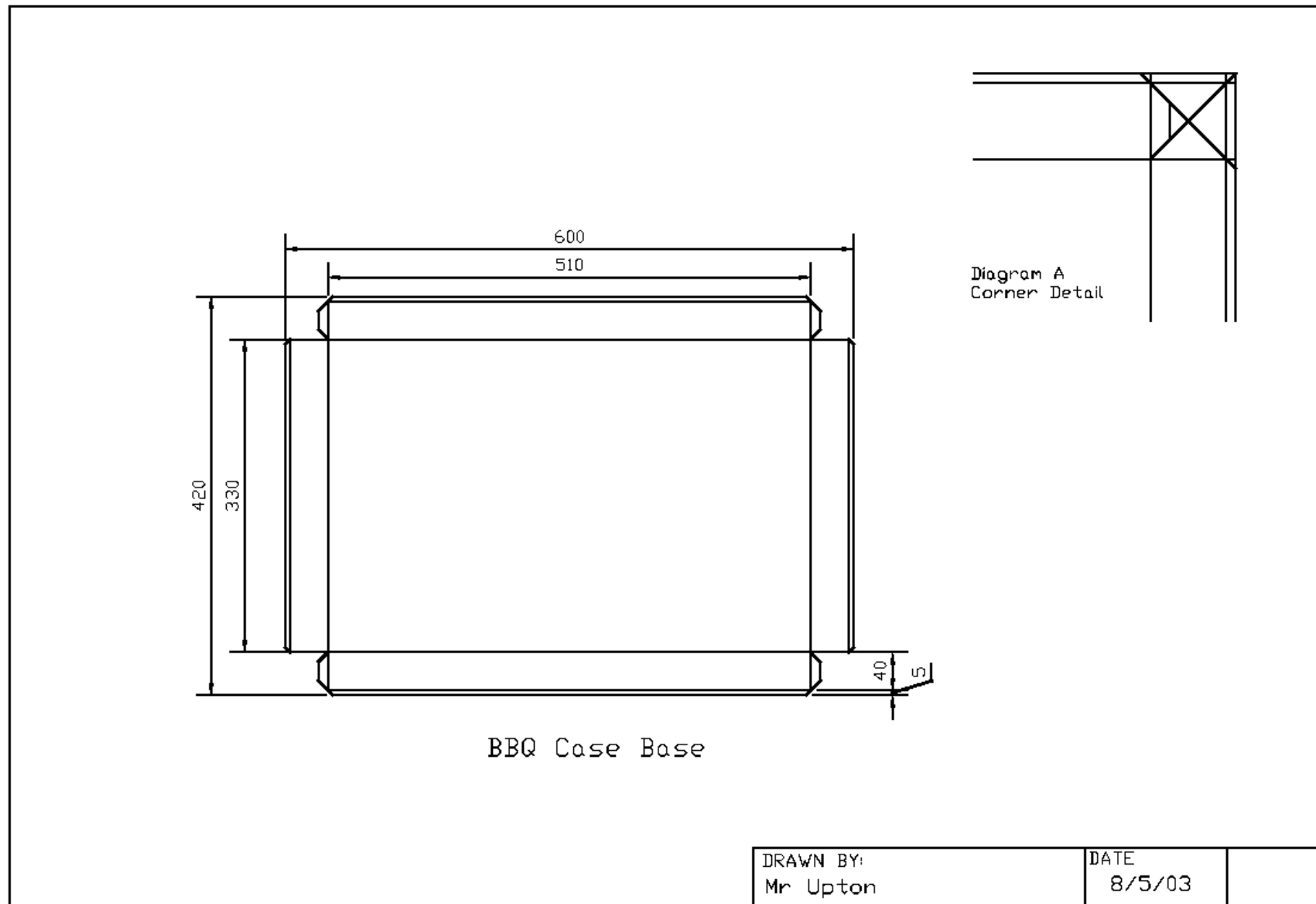
I have completed the theory, checks and written evidence, then saved or printed it as directed.

Modules 1–3	Modules 4–6	Modules 7–9
☐ Module 1	☐ Module 4	☐ Module 7
☐ Module 2	☐ Module 5	☐ Module 8
☐ Module 3	☐ Module 6	☐ Module 9

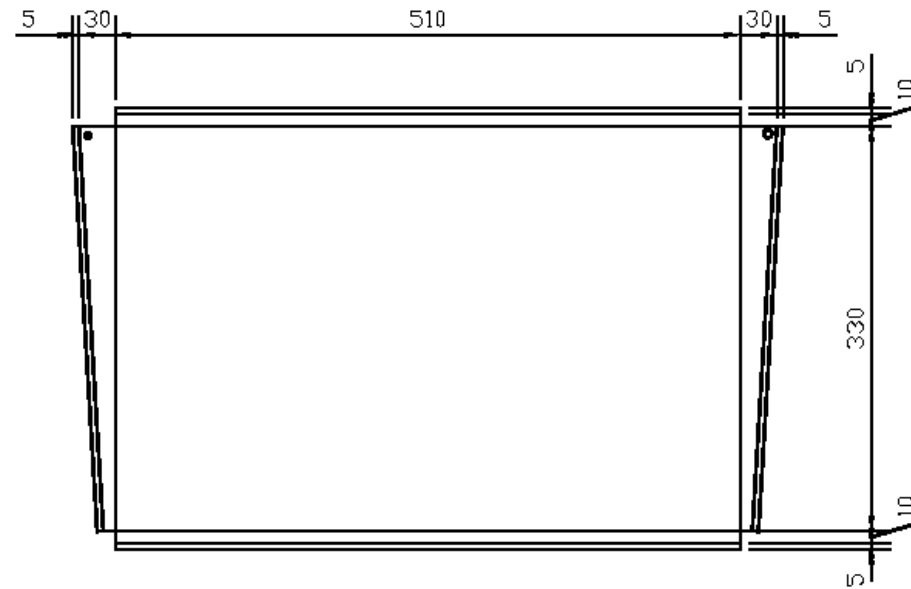
Source course and teacher-led media: stevenowell.github.io/Yr-10-Metal/bbq-case/ . Use the source videos with teacher discussion; the book teaches the core reasoning. Full references, image credits and answers are in the teacher guide. Module sequence does not set current dates or marks.



BBQ · Original source drawing, reproduced complete with black-and-white display for legibility. Confirm the current issued drawing with your teacher. Do not scale from this page.



BBQ Case Base · Original source drawing, reproduced complete with black-and-white display for legibility. Confirm the current issued drawing with your teacher. Do not scale from this page.



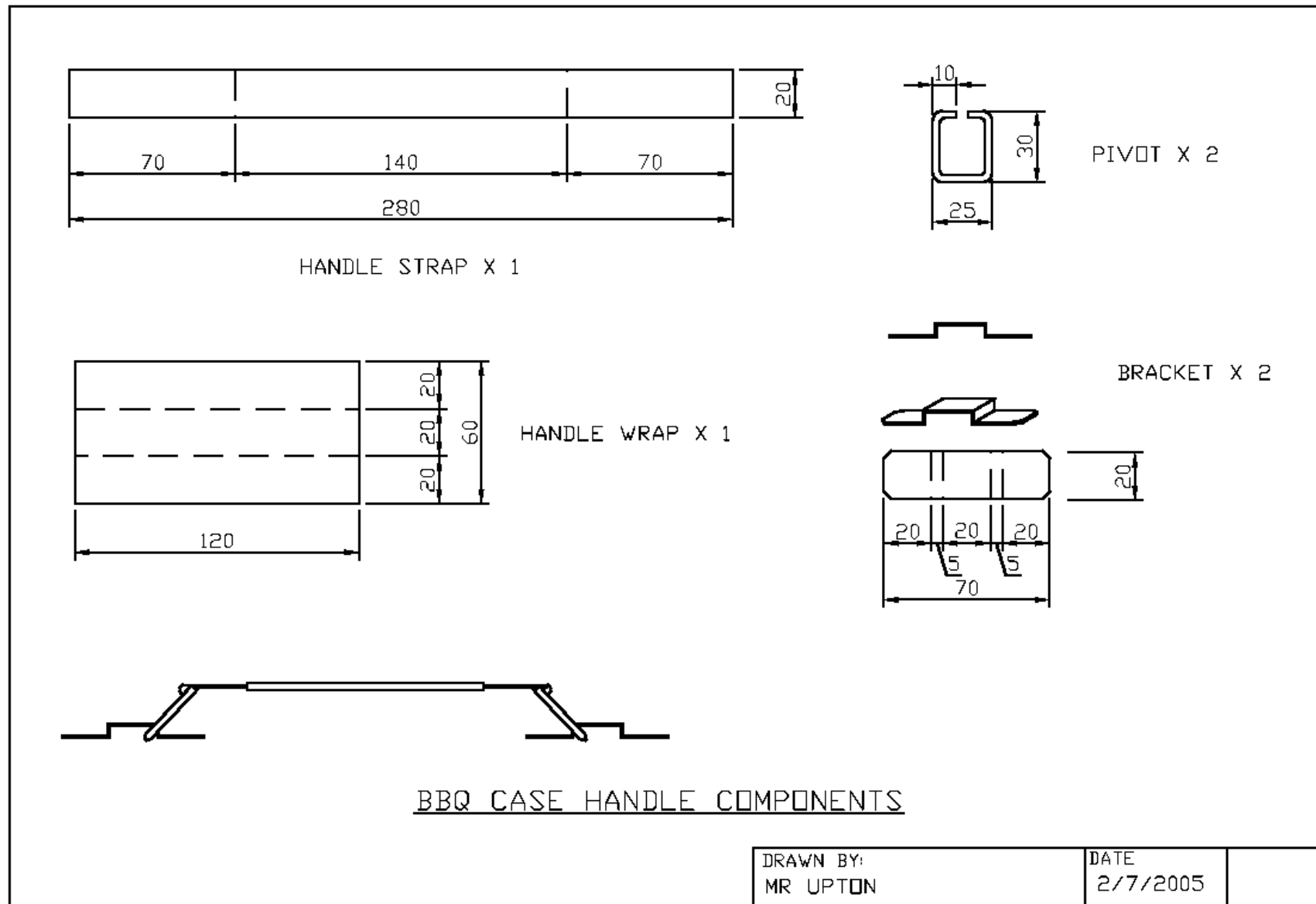
BBQ Case Lid

Note: Calculations for actual lid sizes
 Replace the figures 510 and 330 on
 this plan with the actual outside
 dimensions of your base.

DRAWN BY:
Mr Upton

DATE
1/7/05

BBQ Case Lid · Lid note: replace 510 and 330 with the actual outside dimensions of the completed base. Confirm the current issued drawing and calculation with your teacher; do not scale from this page.



BBQ Case Handle Components · Original source drawing, reproduced complete with black-and-white display for legibility. Confirm the current issued drawing with your teacher. Do not scale from this page.

READ / OBSERVE / APPLY

Plan the product and the evidence

Module 1 / Brief, evidence and materials

1.1 Reading the BBQ-and-case brief as a production contract

Treat the approved brief as a production contract: it defines required results, limits and evidence. Turn the brief, drawings and evidence list into a folio checklist before practical work. A requirement must be met; a preference is possible only where the brief permits flexibility. The BBQ and case form a product system: a decision about one can affect the other's interface, checks or documentation. Confirm unclear details rather than inventing geometry. A source-based checklist connects the intended work, production sequence and final proof.

1.2 From hazard review to a task-specific WMS

A hazard can cause harm; risk considers likelihood and consequence. Apply stronger controls before relying on personal protective equipment (PPE), then review when conditions change. A Standard Operating Procedure (SOP) defines approved operating requirements. A Work Method Statement (WMS) applies them to a specific sequence, identifying tasks, hazards, controls, permission and hold points. The lathe work specifically requires a WMS. A completed document supports planning; it never replaces the current demonstration, supervision or authorisation.

1.3 Inputs, processes and outputs as a planning model

Inputs include materials, drawings, measurements, equipment, time and instructions. Processes transform them; outputs include components, checked surfaces and evidence. A flowchart connects rails, end caps, plate, grills, guard, legs, finish and case instead of merely listing tasks. A quality hold point pauses work to check accuracy, fit, alignment, completeness or permission. If a check fails, a feedback loop returns to investigation or teacher-approved correction. For example, unresolved fit leads back to checking before permanent work, preventing an unnoticed error from moving forward.

INPUT →	PROCESS →	OUTPUT
Approved drawing, identified part and confirmed criterion	Teacher-authorised comparison and recording	Recorded observation or measurement and check result

Worked example: before the next stage, the hold point compares this record with the confirmed criterion. A disagreement returns to teacher review. Work resumes only after the authorised action and required recheck.

Module 1 knowledge check

Questions 1–15 • Circle one answer for each question.

1. According to the theory, the approved project brief should be treated as what?

- ☐ A. A production contract
- ☐ B. A list of optional ideas
- ☐ C. A record completed after production

2. Before practical work begins, what should students create from the brief, drawings and evidence list?

- ☐ A. A final evaluation
- ☐ B. A folio checklist
- ☐ C. A new project design

3. Which example is identified as a project requirement?

- ☐ A. A personal preference for documentation
- ☐ B. An identified function required by the brief
- ☐ C. An unverified idea added during production

4. What should a student do if an important project detail is unclear?

- ☐ A. Invent a practical solution
- ☐ B. Choose the option they prefer
- ☐ C. Confirm the detail with the teacher

5. Which statement best describes a student preference?

- ☐ A. It can override required processes if practical
- ☐ B. It may be considered only when the brief allows flexibility
- ☐ C. It replaces teacher-controlled features

6. What does planning the BBQ and case as a connected product system involve?

- ☐ A. Identifying shared requirements, interfaces and evidence
- ☐ B. Assuming both products use the same construction
- ☐ C. Guessing missing geometry before production

7. A student wants to change a teacher-controlled feature because they think it will work better. Based on the theory, what is the correct response?

- ☐ A. Proceed if the change seems reasonable
- ☐ B. Treat the feature as fixed unless the teacher approves a change
- ☐ C. Replace the feature with a personal preference

8. How does a clear checklist support quality control?

- ☐ A. By replacing the need for evidence
- ☐ B. By showing that production followed the approved brief
- ☐ C. By allowing project requirements to change during production

9. Which planning approach best matches the key takeaways?

- ☐ A. Mix compulsory conditions and personal preferences in one list
- ☐ B. Ignore links between the BBQ and case until evaluation
- ☐ C. Record allowable preferences separately from compulsory conditions

10. According to the source boundary, where should unresolved project-specific details come from?

- ☐ A. The teacher-issued project documents and demonstrations
- ☐ B. Student judgement during production
- ☐ C. General assumptions based on similar projects

11. What is the main purpose of hazard identification during a task?

- ☐ A. To recognise what could cause harm
- ☐ B. To approve the use of equipment
- ☐ C. To describe the operating steps for a machine

12. According to the theory, what is the primary role of an SOP?

- ☐ A. To combine planning, permissions and hold points for a project
- ☐ B. To provide the approved operating requirements for a particular tool, machine or process
- ☐ C. To record evidence that production has been completed

13. Which combination best describes what a Year 10 Work Method Statement (WMS) records?

- ☐ A. Task sequence, hazards, controls, permission requirements and hold points
- ☐ B. Only a list of general workshop hazards
- ☐ C. Only the completed evidence collected after production

14. Why does the theory recommend planning safety decisions before practical work begins?

- ☐ A. To reduce the need for supervision
- ☐ B. To avoid making hurried safety decisions during production
- ☐ C. To replace the need for current demonstrations

15. A project includes cutting, drilling, machining, joining and finishing. What does the WMS help students identify?

- ☐ A. Which different SOPs or demonstrations apply at different stages
- ☐ B. Which project dimensions should be changed
- ☐ C. Which assessment result will be awarded

Module 1 knowledge check

Questions 16–29 • Module 1 question 30 continues on page 11.

16. According to the theory, why do lathe operations require particular attention in the WMS?

- A. Because lathe work is exempt from normal planning
- B. Because lathe operations are specifically identified as requiring a WMS
- C. Because lathe work never requires supervision

17. A student finishes writing a complete WMS. What does the theory say this document provides?

- A. Automatic permission to begin using equipment
- B. A replacement for teacher demonstrations
- C. Evidence that planning has been completed

18. Which statement correctly distinguishes hazard reviews from SOPs?

- A. Hazard reviews identify risks, while SOPs define approved procedures
- B. Hazard reviews replace SOPs once production starts
- C. Hazard reviews and SOPs have exactly the same purpose

19. Which situation best demonstrates the purpose of a hold point in a WMS?

- A. Skipping directly to the next stage once work feels complete
- B. Stopping work at a planned point for checking or teacher authorisation
- C. Recording the final project evaluation after production

20. According to the source boundary, where do unresolved task-specific safety and operating details come from?

- A. The student's previous project notes
- B. The completed WMS alone
- C. The current teacher-issued SOPs, WMS template and demonstrations

21. According to the input-process-output model, which item is an input?

- A. A completed component
- B. Approved materials
- C. Folio evidence

22. What is the main purpose of a process in the input-process-output model?

- A. To transform inputs into products and evidence
- B. To replace teacher instructions
- C. To record only the finished project

23. A student creates a flowchart that simply lists tasks from top to bottom without showing connections or checks. Based on the theory, what is missing?

- A. The project dimensions
- B. The equipment settings
- C. The way stages connect through inputs, changes, checks and evidence

24. What is the purpose of a quality hold point?

- A. To pause production so quality or approval can be confirmed
- B. To increase the number of production stages
- C. To replace the production plan

25. If a quality check identifies a problem, what does the feedback loop show should happen?

- A. Continue to the next stage anyway
- B. Return to an earlier stage for correction or further planning
- C. Start the entire project again

26. Which group contains only examples of inputs mentioned in the theory?

- A. Approved materials, drawings and teacher instructions
- B. Completed components, checked surfaces and folio evidence
- C. Assembled work, completed components and checked surfaces

27. Why does the theory describe the flowchart as a practical control document?

- A. Because it replaces teacher demonstrations
- B. Because it helps coordinate resources, sequence work, document quality and prevent unnoticed errors moving forward
- C. Because it contains all unresolved project-specific details

28. Which statement best describes an output in this planning model?

- A. Information needed before work begins
- B. A resource that controls the exact production sequence
- C. Completed work or evidence produced during production

29. According to the source boundary, where do the exact production sequence and unresolved project-specific details come from?

- A. The teacher-issued production plan, drawings and demonstrations
- B. The student flowchart alone
- C. The input-process-output model itself

READ / OBSERVE / APPLY

From stock to checked setup

Module 2 / Drawings, rails and accurate setup

2.1 Selecting stock form by function and manufacture

Stock selection connects material behaviour, form and manufacture. Rectangular hollow section (RHS) and plate are confirmed categories, but grades, sizes and quantities come from the issued drawing. Consider loads, conditions of use, joining access and the approved equipment. A form may perform its function yet restrict access needed to make a connection. Plan quantities and useful offcuts to reduce waste while keeping every authorised specification.

2.2 Turning drawings into freehand sketches and a spreadsheet cutting list

Read the four coordinated sheets on pages 3–6: BBQ, BBQ Case Base, BBQ Case Lid and BBQ Case Handle Components. Identify titles, dates, views, labels and notes before interpreting a feature. Cross-check orthographic views, the developed base and corner detail, and the separate handle parts against their assembled relationship. Never scale an image to fill a gap.

A freehand sketch explains the form and verified feature relationships; it does not replace the drawing. The spreadsheet records component, stock, confirmed dimensions, quantity, units, source and production notes. Reconcile each row with its sketch and labelled view. Track issue dates and approved revisions. The lid note requires the completed base's actual outside dimensions, so its nominal figures are not unconditional final lid sizes.

2.3 Datums, orientation and quality gates for rails and end caps

A datum is an agreed reference edge, face or point. Consistent references limit cumulative error when related parts must align. Mark orientation, matching parts and the waste side so a correct feature is not processed on the wrong side. At a quality gate, compare the drawing, marks and matching components before irreversible work. A reversible check can expose an error while correction is still possible; record the result and seek teacher direction where uncertain.



Illustrative stock forms. Visible shape does not establish grade, size or suitability. Use current drawing and cutting information.

Modules 1 and 2 knowledge check

Question 30: Module 1 follow-up · Questions 31–45: Module 2.

30. What should each stage of the BBQ and case production flowchart identify before work moves forward?

- A. Only the name of the task
- B. The estimated completion date
- C. What enters the process, what changes, what must be checked and what evidence should be recorded

31. According to the theory, selecting metal stock involves choosing what?

- A. Only a material
- B. A form suited to function and manufacture
- C. Only the lowest-waste option

32. Where must the exact stock grades and sizes be confirmed?

- A. The issued drawing and cutting information
- B. The student's preferred design
- C. General workshop experience

33. Which factor should be considered when judging a suitable stock form?

- A. Personal preference
- B. Workshop wall displays
- C. Joining access

34. Why should students identify usable offcuts before production?

- A. To reduce unnecessary waste through planning
- B. To change the authorised specification
- C. To avoid checking quantities

35. Which statement best matches the theory about manufacturing efficiency?

- A. Reduce waste by changing specifications
- B. Reduce waste through careful planning
- C. Ignore offcuts if enough stock is available

36. What should guide unresolved stock specifications and project details?

- A. Previous student projects
- B. Supplier recommendations
- C. Teacher-issued drawings, cutting information and demonstrations

37. A stock form seems suitable but blocks joining access. What does the theory suggest?

- A. Reassess the stock form because manufacture is affected
- B. Use it anyway because function matters most
- C. Ignore joining access if waste is low

38. Which combination best reflects the key considerations when selecting stock form?

- A. Colour, cost and appearance
- B. Function, properties and manufacturing needs
- C. Weight, popularity and supplier

39. Which condition could make an otherwise suitable stock form impractical?

- A. It has a familiar appearance
- B. It creates excessive waste
- C. It matches the component's function

40. Why should available workshop processes be considered during stock selection?

- A. To confirm assessment marks
- B. To choose the cheapest option
- C. To ensure the stock can be processed accurately

41. What should students do before marking or cutting material?

- A. Measure the drawing image
- B. Convert drawings into organised production information
- C. Estimate missing component data

42. Why should information be cross-checked across different drawing views?

- A. Different views can reveal different component features
- B. Each view uses different measurement units
- C. One view always contains incorrect information

43. What is the correct response when data appears to be missing from a drawing?

- A. Scale the image carefully
- B. Use a reasonable allowance
- C. Leave it unresolved until clarified

44. What is the main purpose of a freehand component sketch?

- A. Replace the formal drawing
- B. Show how the component has been interpreted
- C. Provide exact manufacturing tolerances

45. Which detail is appropriate to include on a freehand sketch?

- A. Brief annotations linked to verified dimensions or notes
- B. Estimated quantities for production
- C. Dimensions measured from the printed image

Module 2 knowledge check

Questions 46–60 • Circle one answer for each question.

46. Which cutting-list entry should remain blank?

- ☐ A. A confirmed component name
- ☐ B. A written drawing dimension
- ☐ C. An unconfirmed material allowance

47. Which spreadsheet practice best supports accurate production information?

- ☐ A. Mixing units to match stock types
- ☐ B. Using consistent units for every entry
- ☐ C. Rounding written dimensions for convenience

48. What should be recorded when the drawing provides revision information?

- ☐ A. The current revision or issue date
- ☐ B. The student's preferred version
- ☐ C. The date material was collected

49. When should the spreadsheet cutting list be updated?

- ☐ A. Whenever a student notices a possible improvement
- ☐ B. After material has already been cut
- ☐ C. When an authorised drawing change occurs

50. What final check should occur before practical work begins?

- ☐ A. Compare the cutting list only with stock available
- ☐ B. Cross-check drawings, sketches, cutting list and teacher clarification
- ☐ C. Measure every drawing view with a ruler

51. What is a datum used for during marking out?

- ☐ A. A decorative edge
- ☐ B. An agreed reference edge, face or point
- ☐ C. A completed inspection record

52. Why should related components be marked from a consistent datum?

- ☐ A. To limit cumulative marking error
- ☐ B. To increase material removal
- ☐ C. To avoid using drawings

53. Which item can orientation marks identify?

- ☐ A. Surface finish quality
- ☐ B. Matching parts
- ☐ C. Production cost

54. What is one purpose of marking the waste side of a line?

- ☐ A. To record inspection results
- ☐ B. To confirm material grade
- ☐ C. To reduce the chance of processing the wrong side

55. When should a quality gate occur?

- ☐ A. Before irreversible work
- ☐ B. After every component is finished
- ☐ C. Only during final assembly

56. What should students compare at a quality gate?

- ☐ A. The component with the drawing
- ☐ B. The drawing with another class
- ☐ C. The workshop with the production plan

57. Why is staged inspection more reliable than waiting until final assembly?

- ☐ A. It guarantees a perfect result
- ☐ B. It prevents one early error being repeated
- ☐ C. It removes the need for drawings

58. According to the theory, what should also be confirmed at a quality gate besides the drawing?

- ☐ A. Workshop layout
- ☐ B. Orientation and relationship to matching parts
- ☐ C. Material price

59. Which practice best reduces cumulative error when marking related components?

- ☐ A. Using several different reference edges
- ☐ B. Estimating positions visually
- ☐ C. Transferring measurements from one consistent datum

60. Where should unresolved rail, end-cap and inspection details be confirmed?

- ☐ A. The teacher-issued drawing, production plan and demonstration
- ☐ B. Student assumptions from previous projects
- ☐ C. Measurements scaled from printed images

READ / OBSERVE / APPLY

Control a drilled feature

Module 3 / Joining and controlled assembly

3.1 Hole location is a design feature, not just a mark

A hole is a designed feature: its location affects fit, alignment and later function. Identify the component, correct drawing view and written information before transferring the centre. A consistent datum and orientation marks keep the intended face and location connected to the source. Changing references can shift a correct-looking mark. Before drilling, compare the part, datum, marked centre and matching identity with the drawing. Record where the location came from and what it is intended to do; appearance cannot supply missing information.

3.2 Drill-press principles behind the teacher-issued SOP

The drill-press SOP controls the exact actions, but each requirement has a purpose. Authorisation establishes readiness of the operator, workpiece and task under supervision. Workholding resists movement or rotation. Guarding and tool-to-work clearance separate moving parts and prevent unwanted contact. Controlled feed manages cutting rather than forcing it; swarf affects visibility, cutting and inspection. Stopping is a deliberate stage: the machine must reach the required condition before inspection. Learn the reasons while following the current SOP and demonstration for every method.

3.3 Inspecting drilled features and responding to faults

Inspect against the approved drawing or criterion, not a general impression. Observe location, size, surface, distortion or damage and record useful measurements or photographs. A rough rim describes a condition; it does not prove the cause. Review the drawing, datum, orientation and approved setup evidence before proposing an explanation. Several causes can produce a similar symptom. Record a non-conforming result and ask the teacher before rework, because an improvised correction may remove evidence or damage later fit.



Illustrative hole rims: describe the visible condition. Appearance does not establish location, size, cause or acceptance.

Module 3 knowledge check

Questions 61–75 • Circle one answer for each question.

61. Why is a drilled hole treated as a designed feature?

- ☐ A. Its location affects fit, alignment and function
- ☐ B. It always improves the component's appearance
- ☐ C. It removes unnecessary material

62. Where should the centre position for a hole be traced from?

- ☐ A. An unverified sample part
- ☐ B. The teacher-issued drawing
- ☐ C. The nearest visible edge

63. What should students confirm before drilling begins?

- ☐ A. The correct drawing view, written information and component identity
- ☐ B. The fastest drilling sequence
- ☐ C. The appearance of similar components

64. What is the main purpose of a datum when locating a hole?

- ☐ A. To identify the drill bit
- ☐ B. To decorate the workpiece
- ☐ C. To provide the reference for centre locations

65. What can happen if measurements are taken from changing edges?

- ☐ A. Small errors can shift the feature
- ☐ B. The hole automatically becomes stronger
- ☐ C. The drawing no longer needs checking

66. Why are orientation marks important before drilling?

- ☐ A. They increase drilling speed
- ☐ B. They confirm the correct face, side or end is being processed
- ☐ C. They replace the need for a datum

67. A hole intended to align with another component should remain connected to what?

- ☐ A. Its design purpose
- ☐ B. Its surface finish
- ☐ C. Its manufacturing cost

68. Before drilling becomes irreversible, which check should be completed?

- ☐ A. Compare the workpiece with the drawing, verify the datum, confirm the marked centre and check the part label
- ☐ B. Estimate the centre position by eye
- ☐ C. Copy the hole position from another component

69. Why should students be able to explain where a hole location came from?

- ☐ A. To justify changing the drawing
- ☐ B. To show it was traced to verified design information
- ☐ C. To avoid checking the marked centre

70. Why is recording the final pre-drilling check recommended?

- ☐ A. It proves the hole was drilled first
- ☐ B. It provides evidence that accuracy was planned and controlled
- ☐ C. It replaces the need to compare with the drawing

71. What does authorisation confirm before drill-press work begins?

- ☐ A. The operator, workpiece and planned task are ready under required supervision
- ☐ B. The hole location has already been drilled
- ☐ C. The component no longer requires inspection

72. According to the theory, what is the main purpose of workholding?

- ☐ A. To improve the appearance of the component
- ☐ B. To prevent the component moving or rotating during drilling
- ☐ C. To identify the correct drawing view

73. Why are guarding and tool-to-work clearance considered together?

- ☐ A. They reduce material waste
- ☐ B. They replace the need for supervision
- ☐ C. They help maintain separation from moving parts and avoid unwanted contact

74. What is the purpose of controlled feed during drilling?

- ☐ A. To manage the cutting action instead of forcing it
- ☐ B. To replace inspection after drilling
- ☐ C. To determine the component identity

75. Why must the operator remain aware of swarf?

- ☐ A. It identifies the correct datum
- ☐ B. It can affect visibility, cutting quality and inspection
- ☐ C. It confirms the drawing is accurate

Module 3 knowledge check

Questions 76–90 • Circle one answer for each question.

76. Where must the exact methods for managing drilling issues come from?

- ☐ A. Previous workshop experience
- ☐ B. General machining knowledge
- ☐ C. The current SOP and teacher demonstration

77. How does the theory describe stopping the drill press?

- ☐ A. As a deliberate stage before inspection
- ☐ B. As the point where the drawing is selected
- ☐ C. As an optional pause during marking out

78. After stopping, what does inspection check?

- ☐ A. Only whether the machine is clean
- ☐ B. Whether the drilled feature matches the approved drawing and suits the next stage
- ☐ C. Whether the operator remembers the procedure

79. What does understanding drill-press principles NOT replace?

- ☐ A. Knowledge of design purpose
- ☐ B. Authorisation, supervision and the demonstrated procedure
- ☐ C. Inspection of completed work

80. According to the key takeaways, what should students rely on during drill-press work?

- ☐ A. Memory from previous projects
- ☐ B. General workshop knowledge
- ☐ C. The current SOP rather than memory or general knowledge

81. What is inspection mainly comparing after drilling?

- ☐ A. The completed feature against the approved drawing or quality criterion
- ☐ B. The component against similar student work
- ☐ C. The drilling time against the production plan

82. Which observation best supports that a drilled feature is ready for later production stages?

- ☐ A. It has a clean appearance
- ☐ B. It is suitable for later fitting or assembly
- ☐ C. It matches a previous component

83. Which evidence could be used during inspection according to the theory?

- ☐ A. Recorded measurements
- ☐ B. Personal opinion
- ☐ C. Estimated dimensions

84. A drilled feature is in the wrong location. What does this observation identify first?

- ☐ A. The exact cause
- ☐ B. A fault category
- ☐ C. The required rework method

85. Before suggesting a correction, what should students do?

- ☐ A. Investigate the likely cause using available evidence
- ☐ B. Begin rework immediately
- ☐ C. Ignore earlier production records

86. Which item may help investigate the cause of a drilling fault?

- ☐ A. The demonstrated process
- ☐ B. A personal preference
- ☐ C. The finished assembly only

87. Why should assumptions not be presented as conclusions when investigating faults?

- ☐ A. Because several causes may produce a similar fault
- ☐ B. Because every fault has only one possible cause
- ☐ C. Because appearance always identifies the cause

88. What should happen before any rework begins on a non-conforming feature?

- ☐ A. The feature should be corrected immediately
- ☐ B. The feature should be recorded and referred to the teacher
- ☐ C. The evidence should be removed

89. Why can unauthorised correction be a problem?

- ☐ A. It may remove evidence or worsen the fault
- ☐ B. It always guarantees a better result
- ☐ C. It automatically proves the original cause

90. What does careful inspection and reporting demonstrate during production?

- ☐ A. Quality was controlled throughout production
- ☐ B. Final assembly corrects all earlier faults
- ☐ C. Appearance is more important than evidence

READ / OBSERVE / APPLY

Trace machining from plan to result

Module 4 / WMS and repeatable work

4.1 Reading a lathe operation through the WMS

A lathe WMS connects the drawing requirement, authorised workpiece and intended feature to the approved setup and process. Each stage identifies hazards, controls, permission and inspection hold points. Permission applies to continuing as well as starting: a changed workpiece, drawing, machine, setup or workshop condition requires stopping and review. A WMS is a traceable plan, not an automatically valid permission slip. Use it together with the current lathe SOP, demonstration and supervision.

4.2 Cutting action, reference surfaces and dimensional control

The workpiece rotates while controlled relative movement between tool and material creates the cutting action. Removal is progressive: every change affects remaining shape, dimensions and surface. Keep later measurements linked to a verified datum or reference surface. Changing references can introduce cumulative error. At approved hold points, pause and compare the developing part with written drawing information before more material is removed. Because removed material cannot simply be replaced, early checks protect the remaining work. A smooth appearance alone cannot confirm dimensional accuracy.

4.3 Surface quality, fit and documented teacher review

Judge size, form, surface condition and functional fit together. Size concerns verified measurements; form concerns shape and consistency; surface concerns irregularities; fit concerns the feature's role in the assembly. A smooth but undersized part can still be unsuitable, just as a measured size does not excuse damage. Record observations, measurements and uncertainty honestly. Keep doubtful work separate from accepted components, and obtain teacher review before acceptance or correction. The record must distinguish the observation, judgement and confirmed next action.



Illustrative turned samples: surface appearance must be checked alongside size, form and fit. The image supplies no acceptance criteria.

Module 4 knowledge check

Questions 91–105 • Circle one answer for each question.

91. What is the main purpose of a lathe Work Method Statement according to the theory?

- ☐ A. Connect the drawing requirement to the approved practical process
- ☐ B. Replace the current lathe SOP during machining
- ☐ C. Record final marks after production

92. Which item should a WMS identify before machining begins?

- ☐ A. The final inspection grade
- ☐ B. The teacher-approved setup
- ☐ C. The machine maintenance history

93. Why are hold points included in a lathe WMS?

- ☐ A. To estimate production time
- ☐ B. To reduce material costs
- ☐ C. To check work before irreversible stages continue

94. Which information is organised into controlled stages within the WMS?

- ☐ A. Planned tasks, hazards, controls, permissions and hold points
- ☐ B. Replacement parts and maintenance records
- ☐ C. Assessment marks and attendance details

95. Teacher approval for continuing a task is based on what?

- ☐ A. How long the task has taken
- ☐ B. The demonstrated process, current setup and present conditions
- ☐ C. Whether the workpiece appears acceptable

96. What should a student do if the approved setup changes during the task?

- ☐ A. Continue if the drawing is unchanged
- ☐ B. Adjust the process without reporting it
- ☐ C. Stop and report the change

97. What does traceability mean in the context of the WMS?

- ☐ A. Linking the drawing to the completed work through the approved process
- ☐ B. Remembering each step without written records
- ☐ C. Comparing one student's work with another's

98. When should the WMS be reviewed?

- ☐ A. Only after the final inspection
- ☐ B. When the workpiece, drawing, equipment, setup or conditions change
- ☐ C. Only at the start of the school year

99. Which document does a completed WMS never replace?

- ☐ A. The approved drawing
- ☐ B. Quality evidence
- ☐ C. The current lathe SOP

100. According to the key takeaways, what should students do before irreversible machining stages continue?

- ☐ A. Use hold points for inspection
- ☐ B. Estimate the remaining machining time
- ☐ C. Rewrite the drawing

101. Why should material be removed progressively during lathe work?

- ☐ A. To preserve shape, dimensions and surface condition
- ☐ B. To reduce the need for drawings
- ☐ C. To make the workpiece rotate faster

102. According to the theory, what provides a consistent basis for locating and checking features?

- ☐ A. Surface appearance
- ☐ B. Reference surfaces
- ☐ C. Tool position alone

103. What is the main risk of changing references without a verified reason?

- ☐ A. Improved surface finish
- ☐ B. Reduced machining time
- ☐ C. Cumulative error and difficult measurement comparison

104. Why are planned hold points included during machining?

- ☐ A. To compare the developing component with verified drawing information
- ☐ B. To avoid using measuring equipment
- ☐ C. To make the workpiece look smoother

105. Why are early dimensional checks more useful than finding a fault at the end?

- ☐ A. They guarantee a better surface finish
- ☐ B. Removed material cannot simply be replaced
- ☐ C. They eliminate the need for final inspection

Module 4 knowledge check

Questions 106–120 • Circle one answer for each question.

106. What should final acceptance be based on?

- ☐ A. The component's overall appearance
- ☐ B. How quickly machining was completed
- ☐ C. Verified measurements, the approved drawing and teacher-controlled criteria

107. Which statement best describes machining in this theory section?

- ☐ A. A planned transformation
- ☐ B. Cutting until the part looks correct
- ☐ C. Removing as much material as possible

108. What is the purpose of recording dimensional checks?

- ☐ A. To replace the approved drawing
- ☐ B. To provide evidence that quality was deliberately managed
- ☐ C. To reduce the number of hold points

109. What does the theory say about appearance during inspection?

- ☐ A. Appearance alone confirms dimensional accuracy
- ☐ B. Appearance replaces verified measurements
- ☐ C. Appearance can support inspection but cannot confirm accuracy alone

110. When checking a developing component, what should later decisions remain connected to?

- ☐ A. The established datum or reference surface
- ☐ B. The most recently machined surface
- ☐ C. The tool's cutting direction

111. Which combination should be used to assess a machined feature?

- ☐ A. Colour, weight and cost
- ☐ B. Size, form, surface condition and functional fit
- ☐ C. Speed, noise and appearance

112. Why is appearance alone not enough to judge machining quality?

- ☐ A. A smooth-looking feature may still have incorrect size, form or function
- ☐ B. Appearance always matches measurement
- ☐ C. Visible finish determines every quality requirement

113. Which inspection evidence is supported by the theory?

- ☐ A. Personal opinion only
- ☐ B. Memory after machining
- ☐ C. Approved measurements, observations, photographs or annotations

114. What should a student do if a suspected fault is found?

- ☐ A. Hide it until assembly
- ☐ B. Identify it honestly
- ☐ C. Polish it before recording

115. Why should doubtful work be separated from approved components?

- ☐ A. To stop it being accidentally used later
- ☐ B. To make it easier to store
- ☐ C. To improve its surface condition

116. What is required before a component is accepted or rework begins?

- ☐ A. Another student agrees
- ☐ B. The feature is polished
- ☐ C. Teacher review

117. Why should improvised rework be avoided?

- ☐ A. It always improves appearance
- ☐ B. It may remove evidence, alter another feature or worsen the problem
- ☐ C. It reduces machining time

118. What is the purpose of documenting the inspection result and teacher's direction?

- ☐ A. To create traceable evidence of controlled quality decisions
- ☐ B. To replace approved measurements
- ☐ C. To avoid further inspection

119. Which statement best describes functional fit?

- ☐ A. Whether the feature has a shiny finish
- ☐ B. Whether the feature can perform its intended role within the developing assembly
- ☐ C. Whether the feature matches a photograph

120. According to the source boundary, where do unresolved inspection, fit and rework details come from?

- ☐ A. Previous student projects
- ☐ B. General workshop experience
- ☐ C. The teacher-issued drawing, acceptance criteria, machining SOP and demonstration

READ / OBSERVE / APPLY

Make related parts fit together

Module 5 / Costs, flowcharts and terminology

5.1 Planning plate work from verified references

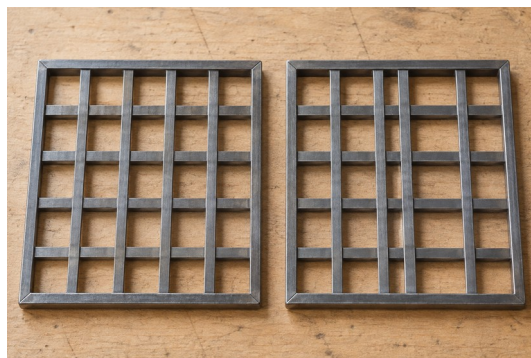
Start plate work with the correct component, all relevant views and notes. Keep its identity traceable through marking and checks. Consistent datums and orientation show how the plate relates to neighbouring parts. Position it within the supplied stock to use sound areas and reduce waste while retaining every authorised requirement. Sequence depends on later relationships: some features need checking before another operation limits access. A hold-point plan records the drawing reference, planned check and expected evidence before irreversible processing.

5.2 Grills and guard: function, spacing and manufacturability

Grills and guard must work as a related set. Repeatability means controlling corresponding features through consistent datums, templates or other demonstrated references. Repeated small variations can affect overall spacing, alignment and fit, so compare with the drawing instead of judging by eye. Manufacturability also includes access for marking, processing, joining, inspection and later cleaning or maintenance. A promising sketch may conceal an access problem. Check related components together before assembly makes correction difficult, and record how consistency was verified.

5.3 Fit-up before permanent joining

Fit-up temporarily brings the plate, grill, guard and supports together for diagnosis before permanent joining. Check identity, orientation, contact, alignment and access against the approved drawing. Consider later parts too: one component may fit now but block the next stage. Restraint can hold the intended relationship while masking poor contact or movement on release. If the approved process uses heat, uneven heating and cooling may affect form or alignment. These are reasons for teacher-controlled methods and review, not instructions to invent a correction. Record and report a mismatch before proceeding.



Illustrative spacing comparison. Use the approved drawing to judge actual gaps and alignment; no project spacing is supplied.

Module 5 knowledge check

Questions 121–135 • Circle one answer for each question.

121. What should planning plate work begin with?

- ☐ A. Verified drawing information
- ☐ B. Material cutting
- ☐ C. Surface finishing

122. Why are consistent datums used when marking a plate?

- ☐ A. To reduce cumulative error
- ☐ B. To increase material thickness
- ☐ C. To replace drawing notes

123. What is the purpose of orientation marks on a plate?

- ☐ A. To improve surface appearance
- ☐ B. To show how the plate relates to surrounding components
- ☐ C. To confirm the material grade

124. Before an irreversible process begins, what should be checked?

- ☐ A. The workshop temperature
- ☐ B. The marking against the drawing
- ☐ C. The production deadline

125. When planning material use, what must not change?

- ☐ A. Verified dimensions or specifications
- ☐ B. The location of existing edges
- ☐ C. The amount of supplied stock

126. Why are planned quality hold points included during production?

- ☐ A. To increase production speed
- ☐ B. To inspect identity, orientation, marking and condition
- ☐ C. To avoid using drawings

127. If sequencing details are unresolved, which source remains authoritative?

- ☐ A. The teacher-issued plan
- ☐ B. Previous student projects
- ☐ C. Personal preference

128. Which action best maintains traceability throughout production?

- ☐ A. Estimating feature locations
- ☐ B. Using clear component identification and recorded checks
- ☐ C. Removing identification after marking

129. When positioning a plate within supplied stock, what should be considered?

- ☐ A. Usable areas, existing edges and other components' needs
- ☐ B. Changing dimensions to fit
- ☐ C. Ignoring remaining stock

130. According to the theory, what should students confirm before marking or processing material?

- ☐ A. The final surface finish
- ☐ B. Any unresolved drawing details
- ☐ C. The machine settings

131. Why should the grills and guard be judged together rather than as separate parts?

- ☐ A. Because their function depends on consistency and alignment
- ☐ B. Because they must always look identical
- ☐ C. Because they are manufactured at the same time

132. According to the theory, what should spacing and alignment be checked against?

- ☐ A. Visual judgement
- ☐ B. The teacher-issued drawing
- ☐ C. The previous component

133. What is the main purpose of using consistent datums, templates or approved references?

- ☐ A. To reduce repeated differences across related components
- ☐ B. To make every component a different size
- ☐ C. To avoid using drawings during production

134. A student spaces grill features by eye because they look evenly arranged. What does the theory recommend instead?

- ☐ A. Accept the result if it appears neat
- ☐ B. Compare only after assembly
- ☐ C. Use verified references from the drawing

135. What does manufacturability include besides producing the design accurately?

- ☐ A. Choosing decorative finishes
- ☐ B. Considering access for processing, inspection and maintenance
- ☐ C. Reducing the number of components

Module 5 knowledge check

Questions 136–150 • Circle one answer for each question.

136. Why can repeated small variations become a problem?

- ☐ A. They can affect overall fit, performance and presentation
- ☐ B. They always improve manufacturability
- ☐ C. They reduce the need for inspection

137. When should related components be compared at quality hold points?

- ☐ A. Only after final assembly
- ☐ B. Before assembly makes correction more difficult
- ☐ C. Only after surface finishing

138. Which evidence can demonstrate that consistency was controlled throughout production?

- ☐ A. A completed assembly alone
- ☐ B. An estimate based on appearance
- ☐ C. Photographs, observations and verified checks

139. A design feature looks suitable in a sketch but blocks inspection access. According to the theory, what does this suggest?

- ☐ A. The feature may be difficult to manufacture or inspect
- ☐ B. The feature should be accepted because it looks correct
- ☐ C. The drawing no longer matters

140. Where must unresolved details about grills, guards and spacing ultimately come from?

- ☐ A. Student judgement
- ☐ B. The teacher-issued drawings, approved quality criteria and demonstrations
- ☐ C. Measurements from a completed BBQ

141. What is the main purpose of fit-up before permanent joining?

- ☐ A. Decorate the assembly
- ☐ B. Diagnose problems while parts can still be separated
- ☐ C. Complete the final joining

142. Which combination is specifically checked during fit-up?

- ☐ A. Orientation, contact, alignment, access and assembly sequence
- ☐ B. Paint colour, surface gloss and packaging
- ☐ C. Workshop timetable, attendance and storage

143. Why should students consider the planned assembly sequence during fit-up?

- ☐ A. It determines the finished appearance
- ☐ B. It may reveal that one correctly fitting part blocks later assembly
- ☐ C. It replaces the need for approved references

144. What is the value of recording photographs and annotated observations during fit-up?

- ☐ A. They provide evidence that component relationships were examined before joining
- ☐ B. They permanently approve the assembly
- ☐ C. They replace the teacher-issued drawing

145. According to the theory, what can restraint do besides holding components together?

- ☐ A. Guarantee perfect alignment
- ☐ B. Prevent the need for inspection
- ☐ C. Mask poor contact or introduce movement when released

146. Where heat is involved in joining, what should students understand?

- ☐ A. Uneven heating and cooling may affect alignment or form
- ☐ B. Heat always improves alignment
- ☐ C. Heat removes the need for fit-up

147. What should happen if a mismatch or uncertainty is found during fit-up?

- ☐ A. Ignore it if the assembly appears close enough
- ☐ B. Report it before permanent joining
- ☐ C. Correct it using any preferred method

148. Which reference should be used when reviewing fit-up?

- ☐ A. Visual appearance alone
- ☐ B. Verified references including the teacher-issued drawing
- ☐ C. An unverified sample component

149. Why should permanent joining not begin simply because the assembly looks correct?

- ☐ A. Appearance alone does not confirm compliance with verified references
- ☐ B. The joining process automatically fixes errors
- ☐ C. Inspection is only needed after joining

150. According to the source boundary, what controls unresolved fit-up and joining details?

- ☐ A. Student preference and previous experience
- ☐ B. General workshop practice
- ☐ C. The teacher-issued drawing, approved joining method and demonstration

READ / OBSERVE / APPLY

Build quality through the assembly

Module 6 / Industry and environmental responsibility

6.1 Stability begins with geometry and matched components

Stable support begins with geometry and matched components, not improvised final adjustment. Consistent datums and orientation marks preserve which faces, edges and ends correspond. Compare the related leg parts before assembly; repeated small variations can accumulate. Rocking, uneven contact or misalignment is a symptom, not a diagnosis. Record it, inspect the relationships and use verified measurements to locate variation. Ask the teacher to review any proposed correction rather than altering a part because it looks responsible.

6.2 Assembly order, access and hold points

Assembly order follows dependencies. Bringing rails, plate, grill, guard and legs together can change access for positioning, inspection or finishing. Before each stage, identify what must already be complete, what needs to remain accessible and what must be checked before it is hidden. Hold points verify identity, orientation, contact, alignment and required evidence. If the teacher changes the sequence, update the flowchart with the reason, approval and effect. A dated revision shows that the new order is controlled.

6.3 Quality assurance during structural assembly

Quality control checks a result at one stage; quality assurance is the system that plans, checks, records and maintains quality throughout the project. Link every observation to its component, drawing requirement, stage and teacher-issued criterion. A useful photograph shows the inspected feature, while its caption explains the check and result. Record a possible non-conformance without hiding it or inventing a cause. Affected work waits for teacher review. Document any authorised rework and repeat inspection so the later judgement has evidence.



Generic matched-pair comparison from one reference. This is not the BBQ leg geometry or an approved correction.

Module 6 knowledge check

Questions 151–165 • Circle one answer for each question.

151. According to the theory, what does stable support begin with?

- ☐ A. Accurate geometry
- ☐ B. Final adjustment
- ☐ C. Extra clamping

152. Why should leg components be prepared from consistent datums?

- ☐ A. To reduce cutting time
- ☐ B. To prevent accumulated variation
- ☐ C. To improve surface finish

153. What is the purpose of orientation marks?

- ☐ A. To identify tool settings
- ☐ B. To reduce waste material
- ☐ C. To preserve the intended relationship between components

154. Before final assembly, what should students compare against the teacher-issued drawing and approved criteria?

- ☐ A. Related components and their geometry
- ☐ B. Workshop layout
- ☐ C. Available measuring tools

155. If an assembly rocks, what does that symptom identify by itself?

- ☐ A. The incorrect orientation
- ☐ B. Nothing about the exact cause
- ☐ C. A damaged component

156. Which of the following is identified as a possible cause of misalignment?

- ☐ A. Paint thickness
- ☐ B. Incorrect orientation
- ☐ C. Tool storage location

157. What should students do first when a stability problem is observed?

- ☐ A. Adjust the assembly immediately
- ☐ B. Request replacement parts
- ☐ C. Record the symptom and inspect the assembly

158. What should be used to locate the source of variation?

- ☐ A. Verified measurements
- ☐ B. Visual estimation
- ☐ C. Previous experience

159. Before making any proposed correction, what does the theory require?

- ☐ A. Reassemble the project twice
- ☐ B. Teacher review
- ☐ C. Additional orientation marks

160. According to the theory, stability should ultimately be achieved through what?

- ☐ A. Repeated adjustment
- ☐ B. Trial and error
- ☐ C. Controlled geometry

161. Why is the approved assembly sequence treated as a production decision rather than just a task list?

- ☐ A. It controls dependencies between components and processes
- ☐ B. It reduces the number of components required
- ☐ C. It guarantees the final appearance

162. Before starting an assembly stage, what should students identify?

- ☐ A. The final project mark
- ☐ B. What must already be complete and what access must remain
- ☐ C. The fastest assembly method

163. According to the theory, assembling a component too early may cause it to

- ☐ A. Improve surface finish automatically
- ☐ B. Reduce the need for inspection
- ☐ C. Restrict later positioning, inspection or processing

164. Why does the theory place a quality hold point before permanent assembly continues?

- ☐ A. Pause production for verified checks before continuing
- ☐ B. Reduce the number of production stages
- ☐ C. Replace the need for approved drawings

165. Which combination is checked at a quality hold point?

- ☐ A. Production speed, cost and appearance
- ☐ B. Component identity, orientation, contact, alignment and required evidence
- ☐ C. Material weight, storage and packaging

Module 6 knowledge check

Questions 166–180 • Circle one answer for each question.

166. What risk does continuing without a hold point create?

- ☐ A. The assembly will always fail
- ☐ B. Later stages become shorter
- ☐ C. An unnoticed error may affect several completed components

167. If the teacher authorises a sequence change, what should be updated?

- ☐ A. The production flowchart and folio record
- ☐ B. Only the completed component
- ☐ C. Only the final photograph

168. Why should students consider inspection and finishing before assembly proceeds?

- ☐ A. To avoid using the approved sequence
- ☐ B. Because later assembly may reduce access to important areas
- ☐ C. To eliminate the need for drawings

169. What maintains traceability after a teacher-approved sequence change?

- ☐ A. Replacing the original drawing
- ☐ B. Restarting the entire project
- ☐ C. Recording the reason, approval and effect of the change

170. Which source controls unresolved sequencing, joining, access and inspection details?

- ☐ A. The teacher-issued assembly plan, drawings and demonstrations
- ☐ B. General workshop experience
- ☐ C. Student preference during assembly

171. What is the main role of quality control during structural assembly?

- ☐ A. Checking results at particular stages
- ☐ B. Planning the entire project system
- ☐ C. Writing the final project report

172. According to the theory, quality assurance is best described as what?

- ☐ A. A final inspection only
- ☐ B. A system showing quality was planned, checked and maintained
- ☐ C. A method for speeding up assembly

173. Which combination makes inspection evidence more traceable?

- ☐ A. An unlabelled final photograph
- ☐ B. A written opinion without evidence
- ☐ C. Photographs with annotations linked to checks

174. Traceable checks should connect each observation to which information?

- ☐ A. The relevant component, drawing requirement, production stage and teacher-issued criterion
- ☐ B. Only the completed assembly
- ☐ C. Only the student's opinion

175. How should a possible failure to meet approved criteria be recorded?

- ☐ A. Ignored until final inspection
- ☐ B. As a non-conformance
- ☐ C. As proof the assembly has failed completely

176. Before affected work continues after a possible non-conformance, what must happen?

- ☐ A. The issue should be hidden
- ☐ B. The work should be repainted
- ☐ C. The teacher should review it

177. What should a non-conformance report avoid doing?

- ☐ A. Claiming an unsupported cause
- ☐ B. Describing the observed issue
- ☐ C. Recording that further review was needed

178. What is required before any rework begins?

- ☐ A. A final photograph
- ☐ B. Teacher authorisation
- ☐ C. Completion of the whole assembly

179. Which information should be included in a documented rework record?

- ☐ A. Only the completed appearance
- ☐ B. Only the date of assembly
- ☐ C. The issue, approved response and later inspection result

180. According to the theory, what does documented rework help prove?

- ☐ A. Assembly quality was built through controlled decisions
- ☐ B. Final appearance alone proves quality
- ☐ C. Inspection records are unnecessary after rework

READ / OBSERVE / APPLY

Prepare the surface, then protect it

Module 7 / Preparation and finishes

7.1 Surface condition controls finish performance

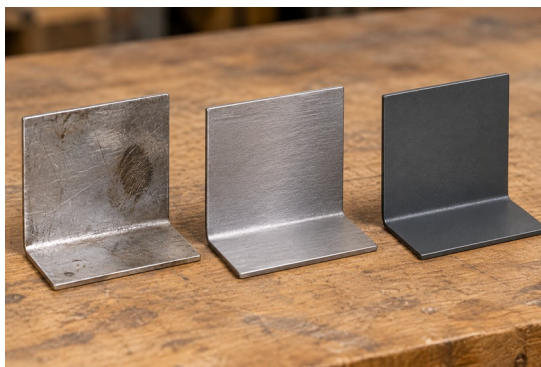
Burrs, scale, contamination, scratches and uneven preparation can affect bonding, appearance and protection. More finish does not repair poor preparation. Preparation establishes the required surface; application places the specified finish on it. Drying reduces wetness or tack, while curing develops the intended properties. A dry-looking surface may not be fully cured. Inspect at approved hold points, record the before-condition and refer uncertain defects to the teacher. Current product information and demonstrated methods control all stages.

7.2 Selecting and applying a finish from authoritative information

Justify the finish through confirmed use conditions, protection, appearance, maintenance and the authorised school process. The product label states intended use and application requirements. The Safety Data Sheet (SDS) addresses hazards, handling, storage and controls; accompanying technical information may give preparation, drying and curing requirements. These sources have different roles. Selection is not permission: a changed product, surface or method needs clarification. Record the actual specified product and the source supporting each criterion. Never infer suitability for heat or food contact from a picture.

7.3 Finish inspection, maintenance and environmental responsibility

Inspect coverage, consistency, defects, edges and corners against teacher-approved criteria. Protect the surface during drying and curing according to current instructions; premature contact or assembly may damage it. Evaluate more than appearance: maintenance, reparability and service life affect future resource demand. Keep claims tied to product information and observed evidence. Manage leftover product, contaminated materials and preparation waste through authorised school procedures. Page 37 records at least two appropriate finishing techniques; two stages of one finish do not automatically count as two techniques.



Illustrative surface conditions. Appearance cannot establish coating identity, adhesion, cure, heat resistance or food-contact suitability.

Module 7 knowledge check

Questions 181–195 • Circle one answer for each question.

181. Why does finish performance begin with surface condition?

- ☐ A. It determines whether the finish can bond consistently
- ☐ B. It shortens the curing process
- ☐ C. It removes the need for inspection

182. According to the theory, applying more finish cannot correct:

- ☐ A. Slow curing
- ☐ B. Poor surface preparation
- ☐ C. Teacher review requirements

183. Which stage produces the required surface condition before any finish is applied?

- ☐ A. Preparation
- ☐ B. Application
- ☐ C. Curing

184. A surface feels dry to the touch. What does the theory say may still be true?

- ☐ A. Preparation is incomplete
- ☐ B. The finish has bonded permanently
- ☐ C. The finish may not yet be fully cured

185. What is the main purpose of planned hold-point inspections before application?

- ☐ A. To identify surface issues requiring review
- ☐ B. To confirm the finish has cured
- ☐ C. To replace teacher instructions

186. Which evidence best records surface condition before application?

- ☐ A. Memory after finishing
- ☐ B. Photographs and annotated observations
- ☐ C. An estimate of drying time

187. Which issue could reduce a finish's intended protection according to the theory?

- ☐ A. Following current product information
- ☐ B. Consistent surface preparation
- ☐ C. Contamination remaining on the surface

188. What should guide every stage of preparation, application, drying and curing?

- ☐ A. Current product information and teacher instructions
- ☐ B. Personal preference from previous projects
- ☐ C. Applying extra preparation whenever possible

189. Which statement correctly distinguishes application from preparation?

- ☐ A. Application creates the required surface condition
- ☐ B. Application places the specified finish onto the prepared surface
- ☐ C. Application begins after curing is complete

190. Why should unverified substitutions or rushed handling be avoided?

- ☐ A. They always reduce drying time
- ☐ B. They guarantee a smoother appearance
- ☐ C. They can affect appearance, adhesion and protection

191. What should primarily justify the selection of a finish?

- ☐ A. Confirmed performance criteria
- ☐ B. Personal colour preference
- ☐ C. The lowest purchase cost

192. Which factor is identified as one consideration when selecting a finish?

- ☐ A. The teacher's favourite product
- ☐ B. Future maintenance requirements
- ☐ C. The number of coats previously used

193. Which source provides authoritative information about hazards, handling and storage?

- ☐ A. The product label
- ☐ B. A completed folio
- ☐ C. The Safety Data Sheet (SDS)

194. Before claiming a finish is suitable, what should students do?

- ☐ A. Check reliable product information
- ☐ B. Compare different paint colours
- ☐ C. Estimate from previous experience

195. What is the main purpose of a product label?

- ☐ A. Record classroom assessment
- ☐ B. Identify intended use and approved application requirements
- ☐ C. Explain all hazard controls in detail

Module 7 knowledge check

Questions 196–210 • Circle one answer for each question.

196. If the planned method differs from what was approved, what should students do?

- ☐ A. Continue carefully
- ☐ B. Rely on memory
- ☐ C. Stop and seek clarification

197. Which combination best supports justified finish selection?

- ☐ A. Label, SDS and technical information
- ☐ B. Appearance and personal opinion
- ☐ C. Past projects and memory

198. What should folio evidence explain about the selected finish?

- ☐ A. Why it was the quickest to apply
- ☐ B. Why it meets the confirmed criteria
- ☐ C. Why it was chosen by other students

199. What controls unresolved handling, application, suitability and curing details?

- ☐ A. General workshop experience
- ☐ B. Internet product reviews
- ☐ C. Teacher-specified product information and demonstration

200. Why should current instructions be consulted and recorded in the folio?

- ☐ A. To create a traceable link between planning, protection and the completed result
- ☐ B. To replace teacher authorisation
- ☐ C. To avoid reading technical information

201. What should finish inspection primarily compare the completed surface against?

- ☐ A. Teacher-approved criteria
- ☐ B. A general impression
- ☐ C. Previous projects

202. Which observation best supports a documented finish inspection?

- ☐ A. The finish looks professional
- ☐ B. Annotated observations identifying inspected areas
- ☐ C. A personal opinion about the colour

203. Why should students inspect edges and corners as well as larger surfaces?

- ☐ A. They are less important than flat areas
- ☐ B. They are easier to photograph
- ☐ C. They may reveal defects or inconsistent coverage

204. What is the main reason a finished surface should be protected while drying and curing?

- ☐ A. To prevent damage before intended properties develop
- ☐ B. To improve the colour automatically
- ☐ C. To reduce the need for inspection

205. Where must the required curing conditions come from?

- ☐ A. General workshop experience
- ☐ B. Current product information and teacher directions
- ☐ C. Student preference

206. When evaluating a finish, what should students consider beyond appearance?

- ☐ A. Only how quickly it was applied
- ☐ B. Only how glossy it appears
- ☐ C. Maintenance, repairability and expected service life

207. Why might a finish needing frequent attention increase resource demands?

- ☐ A. It may require more maintenance or earlier replacement
- ☐ B. It always improves durability
- ☐ C. It eliminates future repairs

208. How should leftover product and contaminated materials be managed?

- ☐ A. Disposed of whenever convenient
- ☐ B. According to teacher directions and authorised school procedures
- ☐ C. Stored without guidance

209. What provides the strongest basis for an honest finish evaluation?

- ☐ A. Personal preference
- ☐ B. Comparison with another student's work
- ☐ C. Observed evidence supported by confirmed product information

210. What should photographs included in finish inspection records demonstrate?

- ☐ A. What was inspected and any uncertainty requiring review
- ☐ B. Only the completed appearance
- ☐ C. Only the best-looking surface areas

LOOK CLOSELY

Develop, form and check the case

Module 8 / Case development and quality

8.1 From three-dimensional case to flat development

A development connects flat faces, edges and joining areas to the formed case. Read it with the current drawings. Reference lines locate information; orientation marks preserve which face and direction belong where. Allowances come from the approved drawing, formula and demonstration, never an estimate from the image. Check while features remain visible and accessible.



Generic base left and lid right. The illustration supplies no clearance, allowance or project dimensions; the real completed base is the measuring source.

8.2 Bending sequence, accumulated error and case fit

Each bend changes later access and geometry. Consistent datums and intermediate checks help identify where an error began. The lid drawing specifically replaces 510 and 330 with the completed base's actual outside dimensions; confirm the measurement and calculation before marking. Springback is partial recovery after forming. Do not guess compensation or change the setup: record unexpected movement or fit and seek teacher direction.

8.3 Hinges and handle as an evidence-based design problem

Define verified constraints before sketching alternatives. Compare movement, access, ergonomics, manufacturability and compatibility with the BBQ and case. An authorised prototype or trial fit may reveal a difficulty that appearance hides. Record evidence and unanswered questions, then justify the selected arrangement. Teacher review and required testing precede any claim that it is suitable or complete.

Explain how a forming hold point helps distinguish marking error, setup error and material response.

Your explanation

Module 8 knowledge check

Questions 211–225 • Circle one answer for each question.

211. What is the main purpose of a development?

- ☐ A. A flat representation of surfaces forming a three-dimensional case
- ☐ B. A finished drawing showing the completed case only
- ☐ C. A photograph of the assembled case

212. Why must a development be read together with the teacher-issued case drawings?

- ☐ A. To estimate missing measurements
- ☐ B. Because development lines may represent different features or production stages
- ☐ C. To replace the approved formulas

213. What is the purpose of reference lines in a development?

- ☐ A. They establish where information is located and transferred
- ☐ B. They decorate the drawing
- ☐ C. They show the finished surface appearance

214. What can happen if orientation marks are not preserved?

- ☐ A. The material becomes thicker
- ☐ B. A correctly shaped area may be formed in the wrong direction
- ☐ C. The approved formulas no longer apply

215. According to the theory, where must allowance values come from?

- ☐ A. Measurements estimated from an image
- ☐ B. Student judgement during forming
- ☐ C. The approved drawing, supplied formula and teacher demonstration

216. Why are allowances included in a development?

- ☐ A. To account for the material and geometry during transformation
- ☐ B. To improve the appearance of the drawing
- ☐ C. To replace reference lines

217. Why are planned hold points included in the production sequence?

- ☐ A. To reduce the number of drawing views
- ☐ B. To confirm verified information before irreversible processing
- ☐ C. To estimate missing allowance values

218. Which statement best reflects the recommended way to read a development?

- ☐ A. As unrelated flat shapes
- ☐ B. Only after forming begins
- ☐ C. As connected surfaces that relate to the completed case

219. Why can later forming make correction more difficult?

- ☐ A. Some features may no longer remain visible or accessible
- ☐ B. Reference lines automatically disappear from the drawing
- ☐ C. Approved formulas become invalid

220. According to the source boundary, who controls unresolved development, allowance, forming and sequencing details?

- ☐ A. Students using prior experience
- ☐ B. The teacher-issued drawings, approved formulas, SOPs and demonstrations
- ☐ C. Photographs of completed cases

221. Why should the teacher-issued bending sequence be followed?

- ☐ A. It changes the final colour
- ☐ B. Early bends affect later geometry, access and alignment
- ☐ C. It guarantees no measurements are needed

222. What is the main purpose of using consistent datums during forming?

- ☐ A. To reduce accumulated error when comparing with the drawing
- ☐ B. To increase the number of bends required
- ☐ C. To avoid using reference lines

223. If checks are taken from changing references, what is the most likely result?

- ☐ A. Improved hardware alignment
- ☐ B. Less need for hold points
- ☐ C. Small differences may accumulate across the case

224. According to the theory, what should be confirmed at planned hold points?

- ☐ A. Orientation, reference lines, developing geometry and trial fit
- ☐ B. Only the finished appearance
- ☐ C. Only hardware positions

225. Which outcome may indicate accumulated error even when no single mistake seems large?

- ☐ A. A smoother surface finish
- ☐ B. Misaligned mating edges or uneven case geometry
- ☐ C. Reduced drawing detail

Module 8 knowledge check

Questions 226–240 • Circle one answer for each question.

226. What does the theory identify as springback?

- ☐ A. A permanent increase in material thickness
- ☐ B. The approved bending sequence
- ☐ C. Partial recovery towards the original shape after forming force is removed

227. How should students respond to unexpected movement or accumulated error?

- ☐ A. Document it and refer it to the teacher before continuing
- ☐ B. Adjust the setup until it seems correct
- ☐ C. Estimate a compensation value and continue

228. Why are intermediate checks completed before another irreversible stage?

- ☐ A. To reduce the number of drawing views
- ☐ B. To identify where variation first appeared
- ☐ C. To confirm the final finish quality

229. What should students avoid doing when springback is suspected?

- ☐ A. Checking the approved process
- ☐ B. Reporting unexpected movement
- ☐ C. Guessing compensation values

230. Which statement best reflects the relationship between each bend and the developing case?

- ☐ A. Each bend can influence later features and references
- ☐ B. Each bend is independent of later stages
- ☐ C. Only the final bend affects case fit

231. What should students do before generating hinge and handle arrangements?

- ☐ A. Define confirmed constraints
- ☐ B. Choose the easiest attachment
- ☐ C. Begin manufacturing immediately

232. How should an unverified assumption about the arrangement be treated?

- ☐ A. As a temporary project fact
- ☐ B. As a question requiring confirmation
- ☐ C. As evidence supporting selection

233. Why should students generate several possible arrangements?

- ☐ A. To avoid documenting decisions
- ☐ B. To guarantee teacher approval
- ☐ C. To compare alternatives before committing

234. Which set contains only approved comparison criteria named in the theory?

- ☐ A. Movement, access and ergonomics
- ☐ B. Cost, colour and popularity
- ☐ C. Speed, weight and assessment marks

235. What is the main purpose of a comparison table or annotated sketches?

- ☐ A. To replace required testing
- ☐ B. To make design reasoning visible
- ☐ C. To prove the arrangement is safe

236. When authorised, why can a prototype provide stronger evidence than appearance alone?

- ☐ A. It confirms every unresolved detail
- ☐ B. It removes the need for approval
- ☐ C. It can reveal practical difficulties

237. What should students record after an approved trial fit?

- ☐ A. Results, problems and unresolved questions
- ☐ B. Only evidence supporting their preferred design
- ☐ C. A claim that the design is complete

238. Which evidence best supports selecting a final arrangement?

- ☐ A. The most attractive sketch
- ☐ B. Approved criteria and testing results
- ☐ C. The first workable idea

239. When may an arrangement be described as safe, suitable or complete?

- ☐ A. After alternatives are sketched
- ☐ B. When the student prefers it
- ☐ C. After teacher review and required testing

240. What does recording alternatives, evidence and approval points demonstrate?

- ☐ A. A controlled design process
- ☐ B. An improvised workshop decision
- ☐ C. A completed manufacturing sequence

READ / OBSERVE / APPLY

Make the record match the result

Module 9 / Problem solving and final evidence

9.1 PMI evaluation that moves beyond opinion

PMI means Plus, Minus and Interesting. Judge the BBQ and case against approved criteria rather than personal likes. A Plus identifies a success with evidence. A Minus identifies a limitation and its effect; any proposed cause remains tentative until supported. Interesting identifies a meaningful relationship or unexpected finding that leads to investigation, not another compliment. Finish with an achievable improvement connected to evidence, likely cause and available processes. An untested proposal is not proof of future success.

9.2 Building the project report as an evidence chain

The report is an evidence chain from brief to evaluation. Link freehand sketches, spreadsheet cutting information, flowchart, SOP evidence and WMS to chronological production records. Caption each image with what it shows, why it matters and the requirement or decision it supports. Explain how a decision led to a quality result and later judgement. Record approved changes and the as-built outcome accurately. Do not invent missing evidence. Current teacher-issued task documents control marks, dates, submission route and unresolved requirements.

9.3 Final quality assurance and authentic submission

Final quality assurance checks both the physical products and the honesty of the folio. Compare the BBQ and case with the confirmed brief and drawings, then check that sketches, lists, flowcharts, WMS, photographs and evaluations tell a consistent story. Identify approved changes, unresolved faults and limitations; do not claim an inspection passed without authorised evidence. Acknowledge sources and assistance without presenting another person's work as your own. Review saved or printed work for completeness, readability and sequence. Readiness is an evidence check, not automatic submission.

REQUIREMENT →	EVIDENCE →	JUDGEMENT / REPORT
Confirmed requirement from the current brief or drawing	Traceable inspection record linked to the production decision	Compare the record with the criterion; explain the supported result and limitation

Illustrative trace: the report connects a production decision to its evidence, judges the result against the requirement and acknowledges limitations. A planned or missing check cannot become a verified result.

Module 9 knowledge check

Questions 241–255 • Circle one answer for each question.

241. What should a PMI evaluation judge the completed BBQ and case against?

- ☐ A. Personal preferences
- ☐ B. The teacher-issued brief and approved criteria
- ☐ C. Other students' projects

242. According to the theory, what should support every Plus observation?

- ☐ A. Evidence such as photographs, measurements, testing records or folio notes
- ☐ B. A personal opinion about quality
- ☐ C. A prediction about future performance

243. When writing a Minus observation, what should students describe before suggesting a likely cause?

- ☐ A. The project's strongest feature
- ☐ B. The effect of the problem
- ☐ C. Another student's approach

244. When should a cause be presented as confirmed?

- ☐ A. Whenever it seems reasonable
- ☐ B. After discussing it with classmates
- ☐ C. Only when supported by available evidence

245. Which statement best describes an Interesting observation?

- ☐ A. It identifies a meaningful result or relationship worth further consideration
- ☐ B. It is another compliment about the finished project
- ☐ C. It lists personal likes about the design

246. A production decision unexpectedly influenced both quality and the relationship between the BBQ and its case. Which PMI category best fits this finding?

- ☐ A. Plus
- ☐ B. Interesting
- ☐ C. Minus

247. What should conclude a strong PMI evaluation?

- ☐ A. A list of favourite features
- ☐ B. A comparison with another project
- ☐ C. One achievable improvement based on evidence

248. Which approach best follows the PMI process?

- ☐ A. Link each observation to approved criteria and evidence
- ☐ B. Record only personal likes and dislikes
- ☐ C. Avoid referring to project evidence

249. Why should proposed improvements be realistic within available processes and project constraints?

- ☐ A. To make the evaluation longer
- ☐ B. So the recommendation can realistically be applied
- ☐ C. To avoid using project evidence

250. According to the source boundary, what controls unresolved judgements, results and improvement claims?

- ☐ A. General workshop experience
- ☐ B. The teacher-issued brief, approved criteria and completed project evidence
- ☐ C. Personal confidence in the conclusion

251. What is the main purpose of the project report?

- ☐ A. Summarise personal opinions
- ☐ B. Form an evidence chain from brief to evaluation
- ☐ C. List every workshop activity

252. Which item should begin the evidence chain?

- ☐ A. The approved brief
- ☐ B. The final evaluation
- ☐ C. Production photographs

253. Why should production photographs be arranged chronologically?

- ☐ A. To reduce the number of images
- ☐ B. To make the report longer
- ☐ C. To show stages, hold points, problems and authorised responses in sequence

254. What should every production photograph include?

- ☐ A. A concise caption explaining its purpose
- ☐ B. Only the date taken
- ☐ C. A decorative border

255. If the completed project differs from the original plan, what should the report do?

- ☐ A. Ignore the difference
- ☐ B. Record the as-built result and acknowledge the change
- ☐ C. Replace the original plan with a new version

Module 9 knowledge check

Questions 256–270 • Circle one answer for each question.

256. Which sources should be identified when they informed decisions?

- ☐ A. Teacher-issued drawings, product information and demonstrations
- ☐ B. Only internet searches
- ☐ C. Only student opinions

257. What should you do if evidence is missing?

- ☐ A. Estimate what probably happened
- ☐ B. Leave it out without comment
- ☐ C. Do not invent it or recreate it inaccurately

258. Until verified, how should dates, teacher details, submission route and component allocations be labelled?

- ☐ A. Teacher to confirm
- ☐ B. Final version
- ☐ C. Approved evidence

259. Which sequence best reflects the intended organisation of the report?

- ☐ A. Evaluation, brief, production
- ☐ B. Brief, production, evaluation
- ☐ C. Production, evaluation, brief

260. According to the source boundary, what controls unresolved task details and evidence requirements?

- ☐ A. Student preference
- ☐ B. Previous project reports
- ☐ C. The current Task 1 handbook, teacher-issued brief, drawings and submission directions

261. What is the main purpose of final quality assurance?

- ☐ A. Increase production speed
- ☐ B. Check both the finished project and its supporting evidence
- ☐ C. Redesign the completed project

262. What should the folio accurately represent?

- ☐ A. What was actually produced
- ☐ B. Only the original design idea
- ☐ C. The best possible outcome

263. If an approved change was made during production, what should happen?

- ☐ A. Remove the original evidence
- ☐ B. Leave the change unmentioned
- ☐ C. Identify the approved change

264. Why should photographs, sketches and quality checks form a consistent account?

- ☐ A. To support an accurate evidence chain
- ☐ B. To improve the project's appearance
- ☐ C. To reduce the number of folio pages

265. What should a student do if a fault remains unresolved?

- ☐ A. Hide it before submission
- ☐ B. Report it honestly
- ☐ C. Assume it passed inspection

266. When may the folio state that a feature passed inspection?

- ☐ A. After an authorised confirmation
- ☐ B. Whenever the student believes it passed
- ☐ C. After comparing it with similar projects

267. Why should teacher-issued drawings, demonstrations and approved references be acknowledged?

- ☐ A. To increase the folio length
- ☐ B. To recognise sources that influenced decisions
- ☐ C. To replace production evidence

268. How should help with planning or production be presented?

- ☐ A. Ignored if only minor
- ☐ B. Described as the student's own work
- ☐ C. Acknowledged honestly

269. Before submission, what should be checked about saved and printed evidence?

- ☐ A. Its completeness, readability and sequence
- ☐ B. Its decorative layout
- ☐ C. Its file size only

270. According to the source boundary, what controls unresolved inspection and submission details?

- ☐ A. The student's personal judgement
- ☐ B. The teacher-issued brief, drawings, criteria, handbook and submission directions
- ☐ C. Examples from previous students

INVESTIGATE / APPLY

Care for tools and act responsibly

Teacher-authorised maintenance and ethical production records

Maintenance preserves a tool's condition and reliable function. Cleaning, lubrication or replacing a consumable may be appropriate, but the teacher confirms the equipment, material and procedure. Record the initial condition, authorised work and observed result. A proposed action is not evidence that it was performed.

Record your teacher-authorised maintenance performance and teacher observation; identify what you actually did.

Maintenance evidence	Actual record
Tool / condition / date	
Teacher-authorised action and source	
Observed result / evidence ID / checkpoint	

Ethical production uses authorised sources, acknowledges assistance and follows the teacher-issued cultural or ethical protocol. Public visibility is not permission to copy cultural knowledge or designs. Identify the appropriate authority before a decision is fixed; never invent consultation, consent or an approval.

Apply the teacher-issued protocol to one actual decision. State any approval or action still pending.

Ethical decision	Individual record
Protocol / context / authority	
Decision applied and reason	
Evidence / acknowledgement / checkpoint	

READ / OBSERVE / APPLY

Follow materials through their life

Resources, Country, composition and complementary materials

Resources connect environmental and social effects

Metal ore is non-renewable: new extraction uses a finite resource. Renewable energy can reduce some processing impacts, but it does not make the ore renewable. Extraction, processing and transport can affect land, water, energy use and people. Longer service life, repair, efficient layouts and recovery keep material useful. Recycling also needs collection, sorting, transport and energy, so it is not impact-free.

Country includes continuing relationships held by Aboriginal and Torres Strait Islander Peoples. Industry can affect landforms, water, access and heritage as well as employment and products. Identify whose Country and Community interests may be affected, engage the appropriate authority early and listen before fixing a decision. A general internet source cannot stand in for that authority or establish consent.

Composition and form answer different questions

A pure metal contains one metallic element; an alloy combines a metal with other elements. Composition and processing can alter strength, hardness, ductility and corrosion behaviour. Brass is a copper–zinc alloy; steel is iron-based. These names do not settle grade, condition or suitability. Compare credible property evidence and applications, then connect the actual specified material to function, manufacture and finish. Stock shape alone is not a pure-metal/alloy comparison.

Secondary materials need a purpose

Ceramics, timber, polymers and textiles can complement a metal product where their properties meet a defined need. A possible grip, protection or insulating role is a concept to investigate, not permission to add it to this BBQ. Compatibility, service conditions, manufacture and end of life matter. Use the approved brief to confirm whether any secondary material belongs in the project.

Explain environmental and societal effects of a material decision, including impacts on Country and Communities. Distinguish non-renewable ore from renewable energy and identify the appropriate authority question.

Apply the material comparison On page 42, compare a pure metal and an alloy using credible evidence, then consider one secondary material's possible role. Keep that investigation separate from the teacher-specified project selection.

INVESTIGATE / APPLY

Compare methods, then record the work

Joining range and traditional / digital production evidence

A joint must suit function, access and later disassembly. Permanent methods are not intended for ordinary separation; semi-permanent fastening can permit removal. Soldering, welding, rivets and fasteners offer different relationships and trade-offs. Heat effects, material compatibility and inspection access matter. The teacher specifies the permitted techniques and practical range; this comparison does not substitute a joining method in the BBQ.

Record distinct authorised joining samples and observations. Compare permanence, disassembly or access; leave unperformed work pending.

Actual technique / source	Purpose, observed result and evidence ID

Traditional and digital technologies both need verified information and quality checks. Digital files can help repeat a feature, but a wrong file can repeat an error. Record the source, approved process, actual contribution and checked result. A saved image of a design is not automatically evidence that a component or surface was made.

Record both teacher-authorised methods, the source or file used, your work, the result and the teacher checkpoint.

Method	Actual component/detailing performance and evidence
Traditional	
Digital	

INVESTIGATE / APPLY

Record two finishing techniques

Use completed products or teacher-authorised samples.

Record two distinct appropriate techniques, each with teacher-observed application. Two stages of one finish do not automatically count as two techniques. Identify the current product/process source and actual evidence. Mark an unperformed technique pending.

Document the before-condition, confirmed preparation process and inspection evidence for one surface. Justify the teacher-specified finish against two confirmed project criteria and cite the source information used.

Technique A / field	Actual record
Technique / product / source	
Before-condition / preparation	
Application / teacher observation	
Quality review / evidence ID	
Criterion 1 / justification	
Criterion 2 / justification	
Technique B / field	Actual record
Technique / product / source	
Before-condition / preparation	
Application / teacher observation	
Quality review / evidence ID	

WORKED EXAMPLE

Connect a decision to its evidence

Industry, enterprise skills and an evaluation example

Enterprise work connects a real need with resources, communication, problem solving and quality. A reliable worker explains a change and its consequences instead of hiding a fault to finish sooner. Decisions also affect users, workers, material use and waste. A technically impressive method adds value only when it suits the job and available resources.

New technology can change tasks and careers. Digital design, controlled production and inspection may create demand for file checking, programming, maintenance and data skills while changing repetitive work. Material knowledge and judgement remain necessary. Compare actual requirements, training and effects; do not assume all jobs disappear or every new process improves every result.



Illustrative case-fit observation. A mismatch is visible evidence; its cause and acceptability need separate checks.

Worked PMI — hypothetical evidence Plus: the trial record shows the lid locating on the base. Minus: one edge is uneven, so fit needs review; the cause is not established. Interesting: the mismatch appeared only when the parts met. Improvement: bring the relevant teacher-approved interface check earlier, then record its result.

The improvement targets an observed problem. In an enterprise, adding an inspection stage also affects time, responsibility and skills. It is a proposal to test, not proof that the fault will vanish. Keep a likely cause tentative until evidence supports it.

Name a new or emerging technology relevant to the example. Cite evidence to judge its career or skill impact; explain an enterprise decision and what still needs verification.

PROJECT FOLIO

Connect the brief to proof

Folio 01 / Project brief and function

Summarise the need, intended user, function and teacher-issued requirements.

Create a short requirements-and-evidence table that links one confirmed brief requirement to the proof the student will collect.

Need, intended user and function: explain the relationship in your own words.

Link each requirement to evidence that could demonstrate it.

Confirmed requirement / source	Proof to collect / when

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Keep requirements and changes traceable

Folio 02 / Requirements and modifications

Separate requirements from constraints and explain any teacher-approved modification.

A requirement states a needed result; a constraint limits the possible solution. Record only confirmed changes.

Category	Requirement, limit or approved change / source
Must achieve	
Constraint or limit	
Approved change / effect	

Prepare a final submission-integrity statement naming verified evidence, approved changes, limitations and source acknowledgements.

Return at completion. Name verified evidence, approved changes, limitations and acknowledgements.

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Trace one operation and annotate the SOP

Folio 03 / WHS and risk controls

Record hazards, risk level, stronger controls, PPE and teacher permission points.

Explain how one teacher-confirmed project operation would be represented in a WMS without copying an SOP verbatim.

Explain how one drawing requirement would be traced through a WMS step, control and inspection record.

Trace field	One confirmed operation
Drawing requirement / step	
Hazard / risk / stronger control / PPE	
Permission / inspection / evidence	

Annotate the teacher-issued SOP by explaining the purpose of one confirmed control and one hold point.

Reference or attach the actual SOP excerpt. Annotate one control and one hold point with their purposes.

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Compare materials and justify selection

Folio 04 / Materials investigation

Compare relevant metals or stock forms and justify the teacher-specified selection.

Justify one teacher-specified stock form using function, manufacturability, availability and finish requirements.

Compare a pure metal and an alloy using property evidence. Investigate one ceramic, timber, polymer or textile role; do not assume it belongs in this project.

Comparison	Evidence and implication
Pure metal / alloy	
Stock form / manufacture	
Possible secondary material	

Justify the actual teacher-specified stock form: function, manufacture, availability and finish.

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Compare two concepts against constraints

Folio 05 / Research and ideas

Record research, freehand sketches or modification ideas and explain the approved direction.

Create a concise decision matrix narrative comparing two student concepts against verified constraints.

Sketch two distinct concepts; annotate the source and verified constraints that each addresses.

CONCEPT A	CONCEPT B

Compare both concepts against the same verified constraints. Explain the approved direction or what still needs confirmation.

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Read two features from the drawings

Folio 06 / Linked evidence tasks 3.1 and 8.1

Explain how the teacher-issued drawing communicates material, dimensions, shape, tolerance or assembly information.

Explain how one confirmed hole feature will be transferred, checked and recorded before drilling.

Annotate a teacher-issued development to explain the role of one reference, one formed feature and one hold point.

Use two distinct source features. Add arrows and explanatory annotations; do not scale measurements from an image.

CASE DEVELOPMENT / REFERENCE	HOLE FEATURE / TRANSFER + CHECK

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. A photo supports your annotations.

PROJECT FOLIO

Reconcile the list, then plan two checks

Folio 07 / Cutting, costing and resources

Summarise your spreadsheet list and explain one decision that reduced waste or improved efficiency.

Describe a verification routine that reconciles a sketch, drawing label and spreadsheet row before cutting.

Name the sketch, label and spreadsheet row. Describe your cross-check and a waste or efficiency decision.

Write a quality-gate checklist for one rail/end-cap stage using only confirmed drawing features.

Stage, confirmed drawing features, checks and evidence / permission:

Rail / end-cap quality gate

Build a short plate-work hold-point plan using verified drawing references and expected evidence.

Drawing reference, stage, hold point and expected evidence:

Plate-work hold-point plan

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Show the sequence and justify the pause

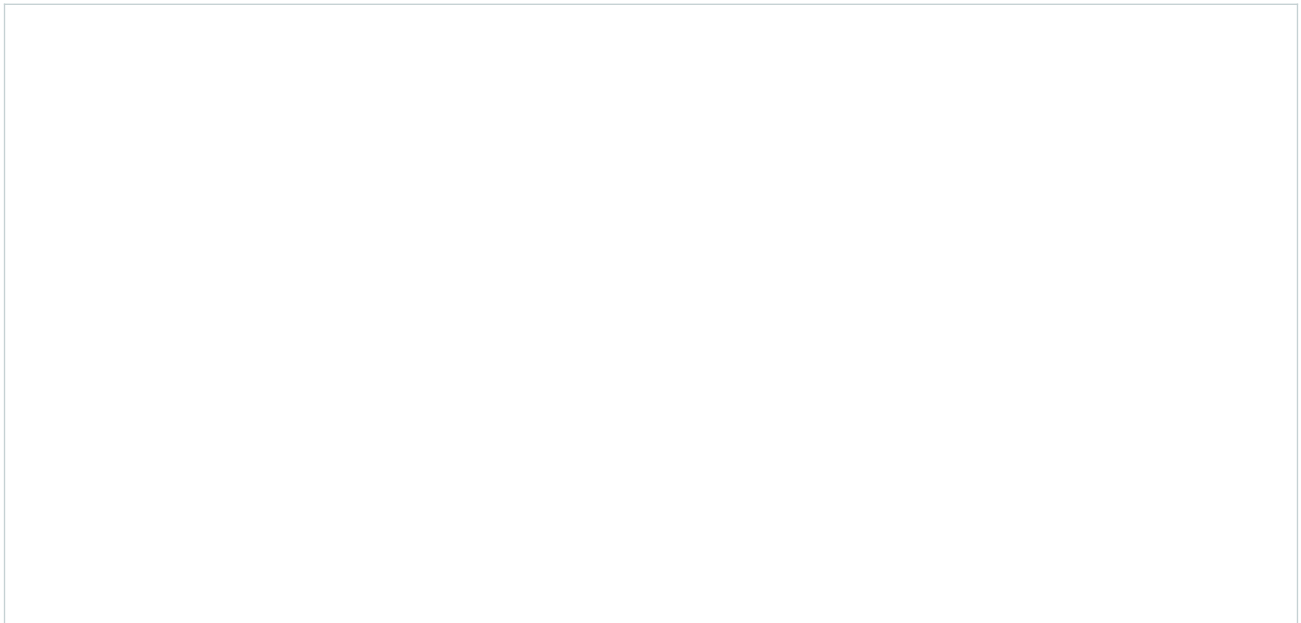
Folio 08 / Production plan and WMS

Sequence key processes, evidence checkpoints, demonstrations and approvals.

Draft one verified branch of the BBQ flowchart and justify where a quality hold point belongs.

Add one justified hold point to the verified production flowchart and explain what evidence is reviewed there.

Draw one verified branch. Label the original hold point and the added or revised hold point; include evidence, demonstrations and approvals.



Justify both hold points. Date the change and state the evidence reviewed at the added point.

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Make production evidence explain a decision

Folio 09 / Production evidence

Write captions for your own photographs that explain the process, control and result shown.

Describe the evidence trail for a supervised machining stage: initial condition, reference, checks and final result.

Write a concise machining quality record separating observation, measurement, judgement and teacher-confirmed next action.

Stage / initial condition / reference / checks / final result; then observation / measurement / judgement / teacher-confirmed next action. Include evidence IDs.

Machining evidence trail

Draft a report paragraph that connects a production decision, evidence, quality result and later evaluation.

Write a connected paragraph using the decision → evidence → quality result → evaluation chain.

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Keep three quality occurrences distinct

Folio 10 / Quality and tolerance checks

Record the approved criteria, measurement or fit evidence, result and any correction.

Record a drilled-feature inspection using criterion, evidence, result and next action.

Criterion / source; evidence / measurement; result; next action:

DRILLING / actual occurrence

Produce a fit-up inspection record with checked criterion, evidence, issue and teacher-confirmed action.

Criterion / source; evidence; issue or result; teacher-confirmed action:

FIT-UP / actual occurrence

Write a quality-assurance entry linking one assembly criterion, its evidence, result and follow-up.

Criterion / source; evidence; result; follow-up and recheck:

ASSEMBLY / actual occurrence

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Compare processes and explain consistency

Folio 11 / Industry and impacts

Compare one classroom and industrial process, then explain a societal or environmental impact.

Name one process in both contexts. Compare scale, control or technology; explain a societal or environmental effect and cite evidence.

Classroom / industrial comparison

Explain how a template, datum or comparison check could support repeatability, subject to the demonstrated process.

Explain how matched-component checks can prevent accumulated error in the leg assembly.

Give separate explanations: repeatability in grills or guard; accumulated-error prevention in matched leg components.

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.

PROJECT FOLIO

Evaluate the evidence and improve the result

Folio 12 / PMI and final evaluation

Use plus, minus and interesting observations to judge function, quality and process, then identify one realistic improvement.

Write one evidence-backed Plus, one Minus with cause, and one Interesting observation leading to further investigation.

PMI	Evidence-backed judgement
Plus	
Minus / cause	
Interesting / investigate	

Evaluate one confirmed finish result using evidence, limitation, maintenance implication and realistic improvement.

Evaluate the finish, then name one realistic improvement. Include evidence, limitation and maintenance implication.

Evidence caption

Source or teacher checkpoint

[] Evidence checked and ready

Optional authorised photo: attach or record the evidence ID. Use your own evidence.